STA 631 • LINEAR REGRESSION

Can Carat Weight Be Determined from Visual Characteristics?

A Linear Regression Analysis of the Diamonds Dataset

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1 Summary of Insights

This project investigates whether the carat weight of a diamond - the primary determinant of its value - can be estimated solely from visually observable characteristics, without physically weighing the stone. Using the diamonds dataset (53,940 observations) from the ggplot2 package in R, we applied linear regression modeling techniques to answer this question.

Research Question

Can carat weight be accurately predicted using only features that a jeweler or buyer can observe visually — specifically physical dimensions (length, width, depth), proportions (table percentage, total depth percentage), and quality grades (cut, color, clarity) — without any direct measurement of mass?

Key Findings

Our analysis revealed three central insights:

- **Physical dimensions are the dominant predictors.**

The length (x) and depth (z) of a diamond, expressed as second-degree orthogonal polynomials, account for nearly all variation in carat weight. This is geometrically intuitive: carat is a measure of mass, and for a material of constant density, mass scales with volume — a non-linear function of linear dimensions.

- **Quality grades add no predictive value once dimensions are known.**

Cut, color, and clarity — while visually observable — were excluded by both best-subset selection and validation-set assessment. Once the physical size of a diamond is known, its grade does not improve the prediction of its weight. Interestingly, there is a well-known market phenomenon where higher-quality grades tend to appear on smaller stones, since large diamonds are difficult to cut and clarity-grade perfectly.

- **A five-predictor polynomial model achieves near-perfect accuracy.**

Our final model — using table, poly(x, 2), and poly(z, 2) as predictors — achieves exceptional fit with a highly parsimonious specification:

Test R^2	Test RMSE	Test MAE	Predictors
0.997	0.026 carats	≈ 0.016 carats	5 (from 25 candidates)

An RMSE of 0.026 carats on held-out test data indicates that typical predictions deviate by only about 26 milligrams - well within any practical tolerance for diamond estimation. The model generalizes exceptionally well, achieving a test R^2 of 0.997 on data never seen during modeling.

Practical Implication

The answer to the research question is a clear yes: carat weight can be determined from visual characteristics with very high accuracy. A jeweler equipped with a millimeter gauge and a table-angle tool - both non-contact instruments - could estimate any diamond's carat weight to within roughly 0.026 carats, without a scale. This has practical relevance for gemological assessment, insurance appraisal, and auction estimation.

2 Modeling Process Summary

2.1 Data Preparation

We loaded the diamonds dataset and began with a structured data quality review. The ordered factors cut, color, and clarity were re-leveled so that the reference category in each case represents the lowest quality grade (Fair, D, and I1 respectively), ensuring that regression coefficients reflect upward deviations from the baseline quality.

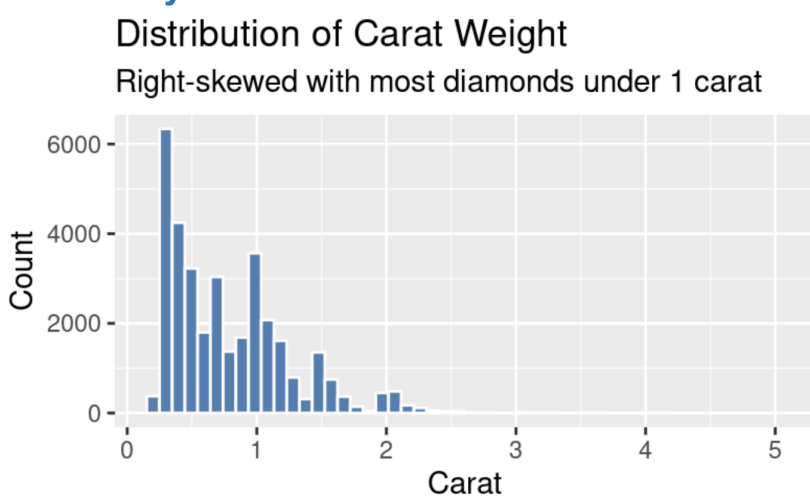
We also identified impossible zero values in the physical dimension variables: 8 zeros in x, 7 in y, and 20 in z. Since a diamond cannot have a zero physical dimension, these represent data entry errors and were removed before any modeling. No missing values were found in any variable.

To prevent data leakage — a critical methodological concern — we implemented a three-way holdout strategy:

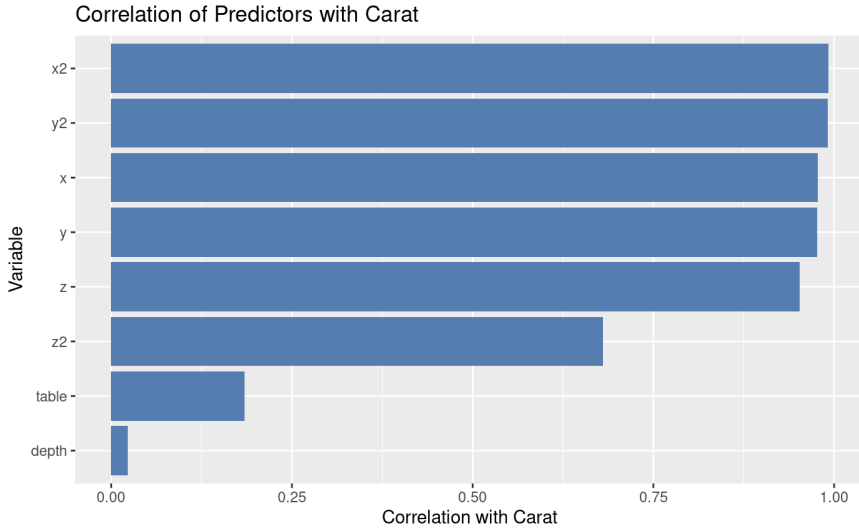
- Training set (64%): used to fit all candidate models
- Validation set (16%): used to compare and select the best model size
- Test set (20%): held out entirely and evaluated only once, at the end

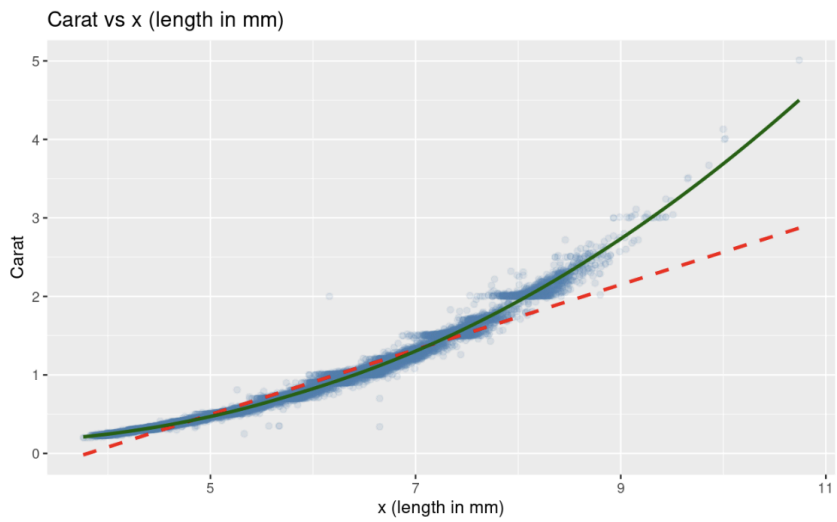
The train–validation–test split ensures that our final performance estimate is unbiased and reflects true generalization to unseen data.

2.2 Exploratory Data Analysis



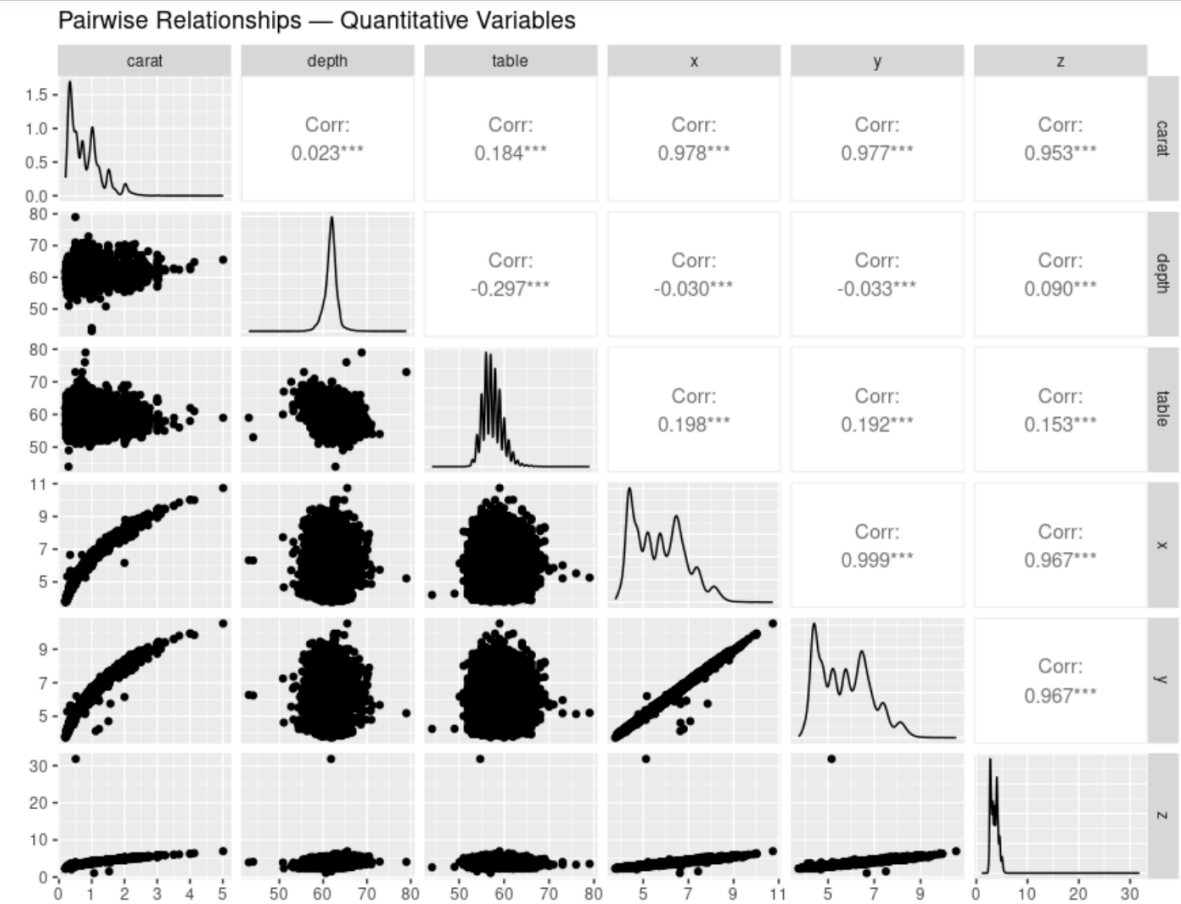
We conducted a systematic EDA to motivate our modeling choices. The response variable (carat) is right-skewed, with a mean of 0.80 and median of 0.70. Most diamonds weigh under 1 carat, with a long tail of larger stones.





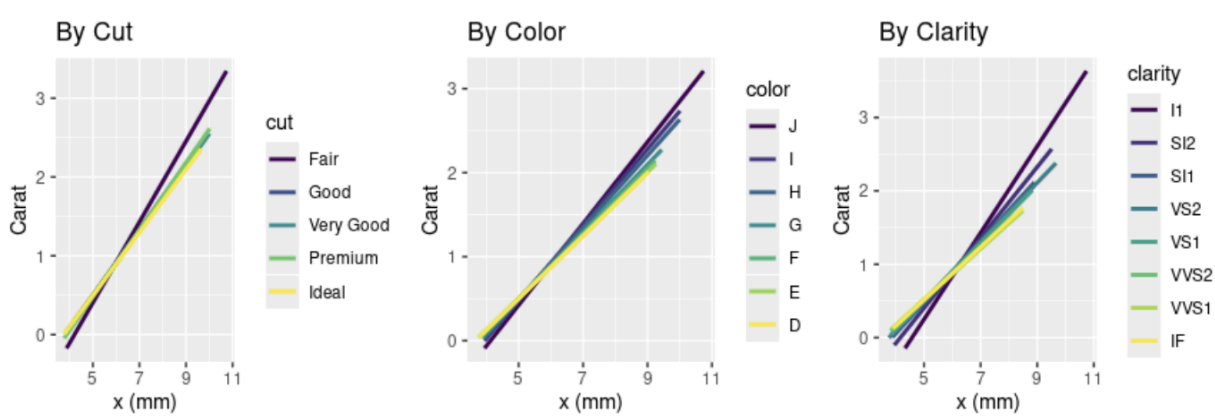
Correlation analysis revealed that the squared dimension terms (x^2, y^2, z^2) were more linearly correlated with carat than the raw dimensions — a direct indication of non-linearity.

A scatterplot of carat vs. x confirmed this: the LOESS smooth clearly deviated from the linear fit, motivating the inclusion of polynomial terms. The red line indicates the linear fit whereas the green line indicates the LOESS (Locally Estimated Scatterplot Smoothing) curve. Instead of one formula for the whole dataset, LOESS curve computes local linear or quadratic regressions at each point, weighting nearby points higher than distant ones. In our visualization, the LOESS curve clearly deviates from the linear fit, confirming that a polynomial term for `x` is warranted.



A pairwise scatterplot matrix confirmed that x, y, and z are highly intercorrelated (all measuring physical size from different orientations), signaling potential multicollinearity. This was addressed in model fitting through the use of orthogonal polynomials.

The low co-relation between `depth` and `carat` indicates that `depth` might not be statistically significant to predict the `carat` of a diamond.



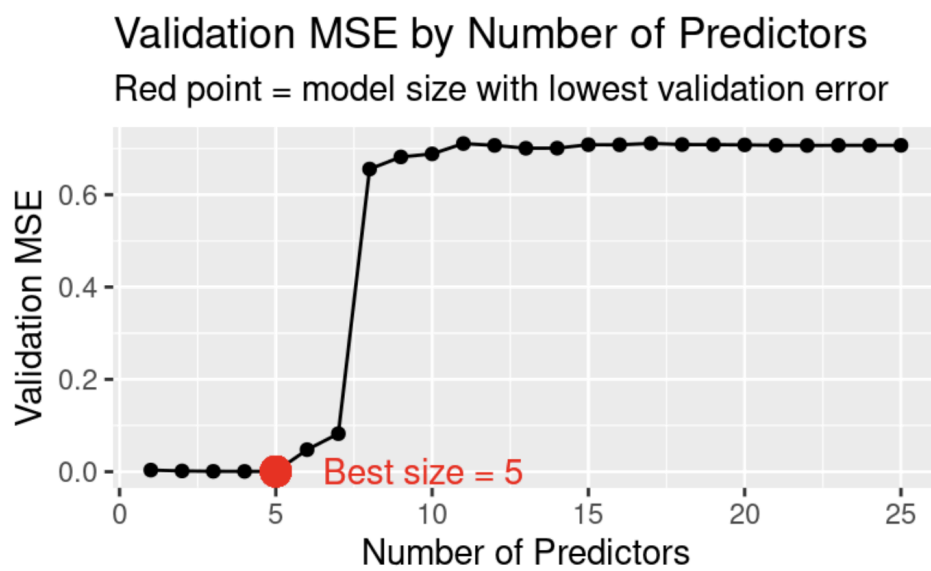
Interaction plots — showing fitted lines for carat vs. x separately across levels of cut, color, and clarity — revealed nearly parallel slopes in all three cases. This indicated that no interaction terms between physical dimensions and quality grades were needed.

2.3 Best Subset Selection

With 25 candidate predictors (including dummy variable expansions of the three factors and squared dimension terms), we applied exhaustive best-subset selection using the regsubsets function from the leaps package. For each model size k (from 1 to 25 predictors), this identifies the single best-fitting model of that size.

We examined four standard in-sample selection criteria: Residual Sum of Squares (RSS), Adjusted R^2, Mallow's Cp, and BIC. These criteria agreed that most of the predictive power is captured within the first 8–15 predictors, with BIC penalizing complexity most aggressively and suggesting around 13.

Rather than relying on these training-set criteria alone, we computed the validation-set MSE for the best model of each size. The validation MSE was minimized at 5 predictors: table, x, z, x^2, and z^2. Crucially, all quality grade variables (cut, color, clarity) were excluded at this optimal size.




```
Best model size (validation): 5
Selected predictors:
 (Intercept)        table            x            z           x2           z2
 0.617548728  0.003185811 -0.700631319  0.462724288  0.074647780 -0.013131996
```

Also, we can see a sudden spike in the MSE value as we go from size 7 to size 8 which indicates that adding an extra predictor has made the model unstable or caused overfitting. In our case, this is likely due to a combination of factors: the inclusion of categorical variables like cut, color, and clarity (which are expanded into multiple dummy variables), and strong multicollinearity among numerical predictors such as x, y, z and their squared terms. When an additional variable is introduced, especially one that is highly correlated or part of a partially included factor, it can significantly distort the model's coefficients and worsen its performance on the validation set.

2.4 Final Model Specification

The selected predictors included both raw and squared versions of x and z — which are inherently collinear with each other. To address this, we replaced them with orthogonal polynomials: poly(x, 2) and poly(z, 2). Orthogonal polynomial encoding captures the same non-linear relationship but eliminates collinearity between the linear and quadratic components.

The final model was fit on the full training data:

$$\hat{carat} = 0.62 + 0.032 \times table + 39.24 \times P1(x) + 18.66 \times P2(x) + 45.93 \times P(z) - 10.22 \times p2(z)$$

where P_1 and P_2 are the orthogonal polynomial components from poly(). All five coefficients were highly significant ($p < 0.001$). The model achieved a training R^2 of 0.998.

Coefficient Interpretations:

1. Intercept (0.615): When all predictors are at their mean values, the predicted *carat* weight is 0.615. Since we're using orthogonal polynomials (which are centered), this is literally the average predicted *carat* in the dataset.

2. *table* (0.00319): For every 1-unit increase in *table* percentage, *carat* weight increases by ~0.003, holding dimensions constant. This is a very small effect — *table* is a minor contributor compared to the physical dimensions.

3. poly(x, 2)1 — 39.24: The linear component of length (x). As the diamond gets longer, *carat* weight increases strongly. The large coefficient reflects how sensitive weight is to even small changes in length.

4. poly(x, 2)2 — 18.66: The curvature component of length (x). This is positive, meaning the relationship between length and *carat* accelerates — longer diamonds gain carat weight faster than a straight line would predict. Think of it as the curve bending upward.

5. poly(z, 2)1 — 45.93: The linear component of depth (z). Actually the largest linear effect in the model — depth is slightly more influential than length in predicting carat weight.

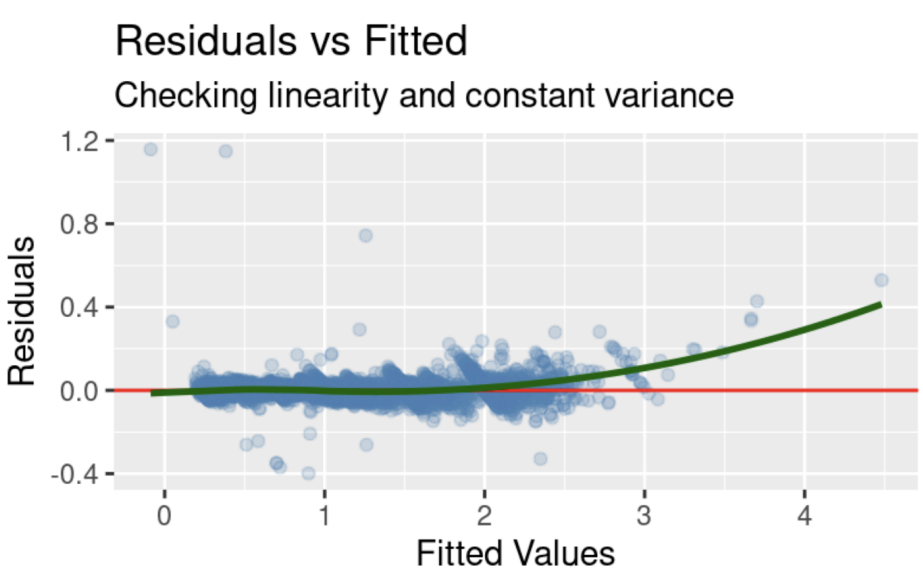
6. poly(z, 2)2 — - 10.22: The curvature component of depth (z). This is negative, meaning the relationship between depth and carat decelerates at larger values — the curve bends downward. Very deep diamonds don't gain as much weight proportionally.

2.5 Diagnostics

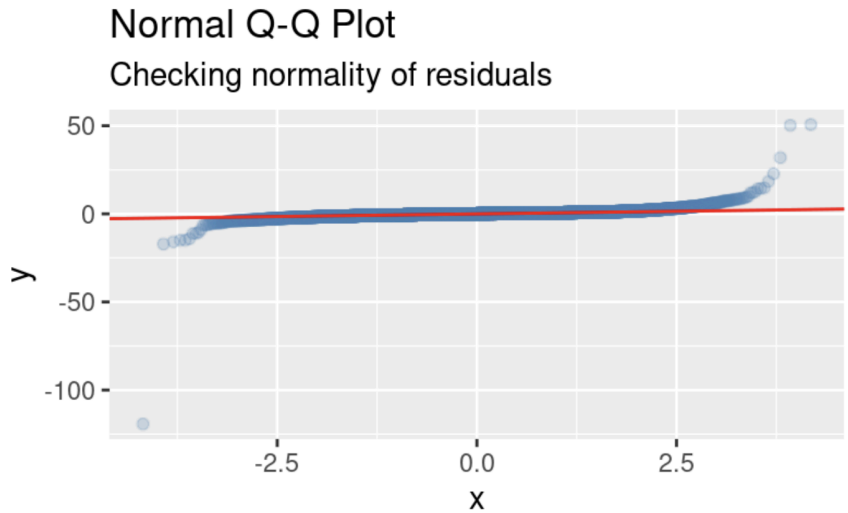
We conducted a full suite of regression diagnostics on the training-set fitted model:

```
                  GVIF Df GVIF^(1/(2*Df))
table        1.144286  1        1.069713
poly(x, 2) 66.527743  2        2.855951
poly(z, 2) 65.550600  2        2.845406
```

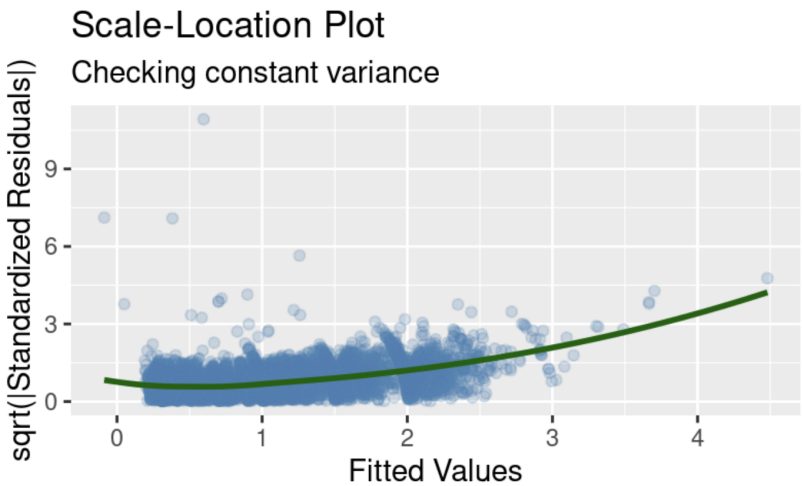
- Variance Inflation Factor (VIF): All GVIF values were within acceptable ranges after switching to orthogonal polynomials, confirming that collinearity was resolved.



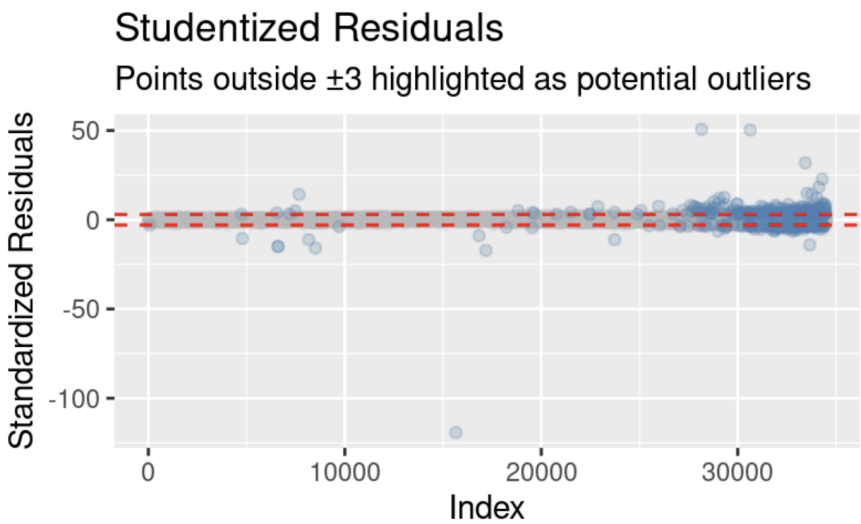
- Residuals vs. Fitted: A slight fan pattern is visible for very large predicted values (> 4 carats), consistent with mild heteroscedasticity at the extremes of the distribution. For the vast majority of diamonds (under 2 carats), variance is approximately constant.



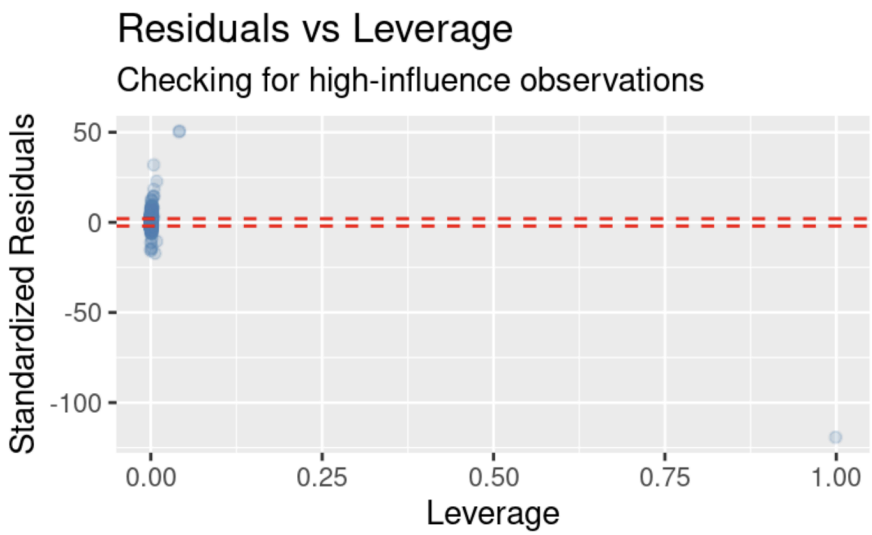
- Normal Q-Q Plot: Residuals are approximately normal in the central range, with mild departures in the tails. This is expected given the right-skewed distribution of carat and the large sample size; central limit theorem effects ensure that inference remains reliable.



- Scale-Location Plot: Consistent with the residuals vs. fitted plot, variance is largely stable except at the high end.



- Outlier Detection: A small number of points exhibit standardized residuals exceeding ±3, likely corresponding to unusual stones. No adjustment was made, as these are real observations and their removal did not materially change results.



- Leverage: No single observation exerts undue influence on the fitted model, given the very large sample size (34,000+ training observations).

2.6 Test Set Evaluation

The test set was held out completely throughout the entire modeling and selection process, and evaluated only once at the very end. This is critical for obtaining an honest performance estimate.

R^2	RMSE	MAE
0.997	**0.026 carats**	**≈ 0.016 carats**

The test performance is essentially identical to validation-set performance, confirming that our model selection process did not overfit to the validation set. A predicted-vs-actual scatter plot showed points falling tightly along the 45-degree line across the full range of carat values, further confirming strong generalization.

2.7 Conclusion

Our final model demonstrates that carat weight can be estimated from visual characteristics with near-perfect accuracy (R^2 = 0.997, RMSE = 0.026 carats). Physical dimensions — specifically length (x) and depth (z), modeled as second-degree orthogonal polynomials — along with the table percentage, are sufficient to explain 99.7% of variance in carat weight. Quality grades (cut, color, clarity), while visually observable, contribute no additional predictive value once dimensions are accounted for. The model is both parsimonious and highly accurate, making it practically useful for non-contact carat estimation.